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SIXTH SEMESTER **B. Tech. (IT)**

END SEMESTER EXAMINATION ***May-2018***

IT306 ARTIFICIAL INTELLIGENCE & EXPERT SYSTEMS

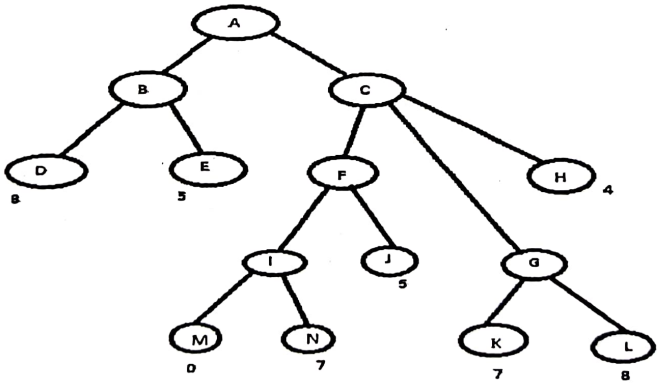
Time: 3:00 Hours *Max. Marks: 40*

Note: Answer any *FIVE* questions. Question no. 1 is compulsory. Assume suitable missing data, if any.

1 Answer all the following questions: **[2x6=12]**

[a] Describe State Space Search example.

[b] Consider the following tree:



In the above tree, the heuristic value of every node is written. Trace the path of the goal L.

[c] Show the operation of resolution using following predicates:

A: P∨Q∨R

B: ~P∨R

C: ~Q

D: ~R

[d] In a database, a rule is defined as:

(P1 AND P2) OR P3 ? R1(0.8) AND R2(0.3)

P.T.O.

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rule with CFs 0.8 and 0.3 respectively. Find the CF of results on the basis of premises.

[e] How AI provide usefulness to Natural Language Processing?

[f] Construct Semantic Net representation for the following:

(i) Pomperian (marcus), Blacksmith(marcus)

(ii) Mary gave the green flowered vase to her favourite cousin.

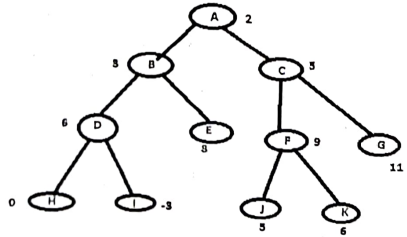
2 Answer all the following questions:

[a] Compare Depth First Search and Hill Climbing Search Procedures. [3]

[b] Explain the production system. What is the meaning of f(n), g(n), h(n)? [4]

3 Answer all the following questions:

[a] Consider the following tree:



The nodes are expanded according to heuristic value written with nodes. Write node expansion sequence for goal I. [3]

[b] Prove that A* is admissible by showing that it fulfills following conditions:

(i) A* search will terminate.

(ii) During its execution, there is always a node on open that lies on an optimal path to the goal.

(iii) If there is a path to the goal, A* will terminate by finding the optimal path. [4]

P.T.O.

4 Answer all the following questions:

[a] What are the merits and demerits of semantic nets and frames? [3]

[b] Assume the following facts:

(i) Ram only likes easy courses.

(ii) Science courses are hard.

(iii) All the courses in the law department are easy.

(iv) Law 301 is a law course.

Use resolution to answer the question:

"What course would "Ram like""? [4]

5 Answer all the following questions:

[a] Explain Genetic Algorithm with an example. [3]

[b] How does rule based Expert System propagate uncertainties? [4]

6 Answer all the following questions:

[a] Write a note on Expert Systems. [3]

[b] With the help of suitable example, explain how the explanation based learning makes use of principle of induction? [4]

7 Answer all the following questions:

[a] Give the general Natural Language Processing(NLP) system and explain its various phases. [3]

[b] Using MYCIN's rules for inexact reasoning, compute CF(Certainty Factor), MB(Measure of Belief), and MD(Measure of Disbelief) of h, when three observations are: [4]

$CF(h_1,o_1) = 0.5$

$CF(h_1,o_2) = 0.3$

$CF(h_1,o_3) = -0.2$

—END—